

Exploring Quantitative Assessment of Cybersickness in Virtual Reality Using EEG Signals and a CNN-LSTM Network

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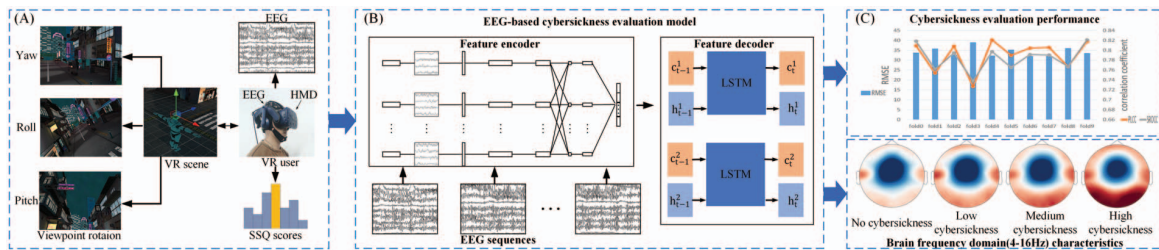
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Figure 1: Overview of our proposed EEG-based framework for quantitative cybersickness assessment and analysis. (a) Experimental design of our proposed cybersickness-induction paradigm with viewpoint rotation when the user is wearing both EEG cap and VR headset. (b) The proposed CNN-LSTM encoder-decoder for quantitating cybersickness based on EEG. (c) The achieved meaningful performance for VR cybersickness evaluation with PLCC of 0.79, SROCC of 0.78, and RMSE of 34.28. (d) The scalp topographies of a frequency band (4-16Hz) which show significant differences in occipital lobe activation.

ABSTRACT

Cybersickness in the virtual reality (VR) environment poses a significant challenge to the user experience. In this paper, we evaluate VR cybersickness quantitatively based on electroencephalography (EEG). We designed a novel experimental paradigm to generate cybersickness based on the virtual rotation of the user's head and collected the corresponding EEG signals. A novel deep learning framework combining a convolutional neural network (CNN) with long short-term memory (LSTM) was proposed to predict cybersickness based on the EEG signals. With the proposed method, a meaningful evaluation of cybersickness is achieved. We further investigated the brain frequency-domain physiological characteristics at different levels of cybersickness.

Index Terms: Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—Heuristic evaluations; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Hardware—Emerging technologies—Biology-related information processing—Neural systems

1 INTRODUCTION

Virtual reality (VR) has attracted attention in various fields such as entertainment, healthcare, and education. However, the VR user experience is often accompanied by cybersickness, which poses a great challenge and safety issue. Cybersickness symptoms include disorientation, oculomotor discomfort, and nausea [6]. In order to handle cybersickness for VR content creation and viewing, it is necessary to quantify cybersickness level induced by VR contents.

Currently, most evaluation methods are mainly based on subjective metrics like questionnaires [2, 7]. However, these subjective methods have several limitations including offline, non-continuous,

non-end-to-end, and easily influenced by individual differences. In physiology, cybersickness is a response to the conflict between vestibular and visual systems. Therefore, various physiological parameters have been widely utilized as the objective indicators to evaluate cybersickness, such as electroencephalography (EEG) [3], electrocardiogram (ECG) [5], galvanic skin response (GSR) [5], and heart rate and blood pressure [8]. EEG is not only an objective, but also a direct reflection of the physiological status of the brain, so it is promising to evaluate cybersickness based on EEG. However, in the current reported studies, the temporal characteristics of EEG have not been considered, and the brain's physiological characteristics under cybersickness still need to be explored.

In this paper, we present a novel EEG-based evaluation method with deep learning to quantize the cybersickness scores of participants in VR. We designed an experimental paradigm with generated VR scenes to induce cybersickness and then collected the EEG signals from 16 participants. Combining convolutional neural network (CNN) with long short-term memory (LSTM), a novel deep learning framework was proposed to train the neural network with the collected EEG dataset and then predict cybersickness scores of the users. And the brain frequency-domain physiological characteristics were investigated under different levels of cybersickness in detail.

2 THE PROPOSED METHOD OF VR CYBERSICKNESS QUANTITATIVE ASSESSMENT

We proposed a CNN-LSTM encoder-decoder framework to automatically learn and extract the time-frequency characteristics of EEG to evaluate cybersickness in VR environment. The details of this framework are described as follows.

EEG preprocessing - A 0.5 Hz high-pass filter was utilized on the recorded EEG data to eliminate the DC drift. After that, the high-frequency noise component was removed from the EEG data using a low-pass filter at 50 Hz. And then the EEG signals were down-sampled to 128 Hz to lower the computational load. Finally, independent component analysis (ICA) was employed to remove artifactual components from the recorded EEG signals.

CNN-LSTM model structure - As shown in Fig. 1(b), this neural network model structure mainly consists of two components:

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CNN-based encoder and LSTM-based decoder. In general, the CNN-based encoder is responsible for feature encoding, which can extract cybersickness-related features from input EEG signals, and the LSTM-based decoder is responsible for decoding the features generated by CNN and outputting the cybersickness score. We designed the CNN-based encoder based on EEGNet [4]. EEGNet is a compact CNN architecture for EEG-based Brain-Computer Interface (BCI) that can be applied across several different BCI paradigms. The LSTM-based decoder mainly consists of two components: a 2-layer LSTM (32 units and 16 units) and a fully connected layer. With the LSTM layer, the decoder could determine the current possible cybersickness states based on the current EEG features and the previous cybersickness states.

Frequency-domain analysis - The characteristics of the EEG data in the frequency domain were extracted. The power spectral densities (PSDs) were estimated using Welch's method. In addition, we located the activated brain regions under different levels of cybersickness using scalp topographies.

3 EXPERIMENTS

A VR experiment was designed for generating cybersickness and collecting the EEG signals. The details of this experiment are described as follows.

Participants - A total of 16 participants (13 men and 3 women with a mean age of 25.6 and a standard deviation of 2.3) participated in the experiment. All participants had normal vision or corrected vision, without a history of neurological or other diseases. The study protocol was approved in advance by the institutional review board.

Experimental paradigm - The experiment investigated four levels of scene rotations: no rotation and viewed scene rotation along the pitch axis, yaw axis, and roll axis. We used the Unity game engine program (Unity Technology) to create the VR content, as shown in Fig. 1(a). The angular velocity of rotation was 30–120°/s with step 30°/s. The condition without rotation was treated as a baseline condition. The length of the one single rotation trial was 1-minute. Participants were asked to wear the HTC VIVE PRO.

Experimental procedure - The first trial was the baseline trial, which comprised a 2-minute static VR scene without rotation. Following that, there were 12 rotational trials total to induce cybersickness (3 rotation axes and 4 angular velocities). After each rotational trial, subjects graded the cybersickness level by completing the SSQ sheet [2] and took a 3-minute rest.

EEG data recording - Brain Products actiCHamp (Brain Products, Germany) was used to record the EEG data at a sampling rate of 1000 Hz. Furthermore, a 27-channel actiCAP manufactured by Brain Products was utilized in our experiments. EEG activity was continuously referenced to the Cz channel.

Model implementation - To ensure performance validity, 10-fold cross-validation was then conducted. We implemented the proposed neural network on the Tensorflow framework [1]. Each input sample is a 1-s EEG segment. We used Adam to optimize the proposed network with a learning rate of 0.001 and a batch size of 16.

4 RESULTS

The results mainly include the performance of the CNN-LSTM model and the physiological characteristics of EEG.

Evaluation performance - Pearson linear correlation coefficient (PLCC), spearman rank order correlation coefficient (SROCC), and root mean square error (RMSE) are utilized as performance evaluation metrics. The collected SSQ scores ranged from 0 to 209.44. Fig. 1(c) shows the individual SSQ prediction performance of the 10-fold cross-validation. The correlation coefficients (PLCC and SROCC) of all folds show significant correlations (with $p < 0.05$). With the results of similar performance across folds, it indicates the generalization performance of the proposed model. The results of all folds were averaged, and the proposed method achieved meaningful

correlation performance with PLCC of 0.79 and SROCC of 0.78 (with $p < 0.05$) and regression performance with RMSE of 34.28.

EEG frequency-domain characteristics - According to the SSQ results of all trials, cybersickness was clustered into three levels by K-means: low(SSQ: 0–40), medium(SSQ: 40–100), and high(SSQ: >100). Generally, in the low-frequency bands, EEG intensity is positively correlated with cybersickness, which indicates that there is a significant statistical correlation between Delta (1.0–4.0 Hz), Theta (4.0–8.0 Hz), and Alpha wave (8.0–12.0 Hz) increasing and cybersickness in VR across participants (with $p < 0.05$). The PSDs show significant differences in the frequency band (4–16Hz) under different levels of cybersickness (with $p < 0.05$). Fig. 1(d) indicates that in the frequency band (4–16Hz), the activation intensity of the occipital lobe is positively correlated with cybersickness (with $p < 0.05$).

5 CONCLUSION AND FUTURE OF WORK

Our work mainly focuses on cybersickness prediction based on the user's EEG data. We proposed a novel CNN-LSTM framework to automatically learn and extract the features related to cybersickness from the EEG. In addition, we investigated the brain frequency-domain physiological characteristics under different levels of cybersickness.

In the future, we will continue to expand the current database and improve the experimental paradigm to make the classification performance of the cybersickness prediction more accurate.

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